

Reconstruction of Clipped Time-History Records, with Application to Ground Motions from the 1992 M_L 5.8 Roermond, Netherlands, Earthquake



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Abstract

Analog-to-digital conversion of time-history data can lead to amplitude truncation (known as clipping) in cases in which a signal exceeds the full-scale range of the digitizer. When digitizers are configured, it is necessary to balance two competing requirements: maximizing instruments' full-scale range for recording large amplitudes, while ensuring weak signals are recorded with sufficient resolution. An example of this trade-off is seen when recording earthquake ground motions, which span several orders of magnitude: from weak-motion teleseisms or local microseismicity, through to large-magnitude earthquakes. This can lead to clipped records in which strong ground motions saturate instruments configured to record weak signals.

To address this problem, we present a new approach: nonstationary clipped waveform restoration, $CWR_{\Delta t}$. The approach is capable of reconstructing even severely clipped waveforms, truncated at as low as 10% of true peak motion. The approach is based on "Projection onto Convex Sets" and builds upon the stationary and deterministic CWR method of Zhang *et al.* (2016). A key advantage of the method presented herein lies with the characterization of temporal uncertainties that can be propagated into other domains such as Fourier, power spectral density, or damped pseudospectral amplitude.

We benchmark the method against artificially clipped ground-motion records from the European Engineering Strong-Motion database. A significant improvement is demonstrated with respect to the original CWR method. Clipped records from the 1992 M_L 5.8 Roermond, Netherlands, earthquake are then reconstructed. This key event in northwestern Europe stands as the region's largest instrumentally recorded earthquake, but from which the majority of records have, until now, been unusable. We successfully verify the $CWR_{\Delta t}$ approach against ground-truth records through comparison with collocated clipped/unclipped records from high- and low-gain instruments at three sites. Finally, we apply the method to clipped records in the epicentral region, providing the tool to develop a key data set for engineering seismology in Europe. The proposed method has a multitude of applications beyond seismology, being generalizable to any time-history data.

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[Supplemental Material](#)

Introduction

High-quality digital records of earthquake-induced ground shaking are an integral part of modern seismology. These data have played a fundamental role in a wealth of scientific and engineering advances, from understanding Earth structure and composition to the quantification and consequent mitigation of seismic risks. Both the instrumentation and the wider networks used to make these records have undergone rapid advances over the last decades. It is easy to forget that as little as 30 yr ago many

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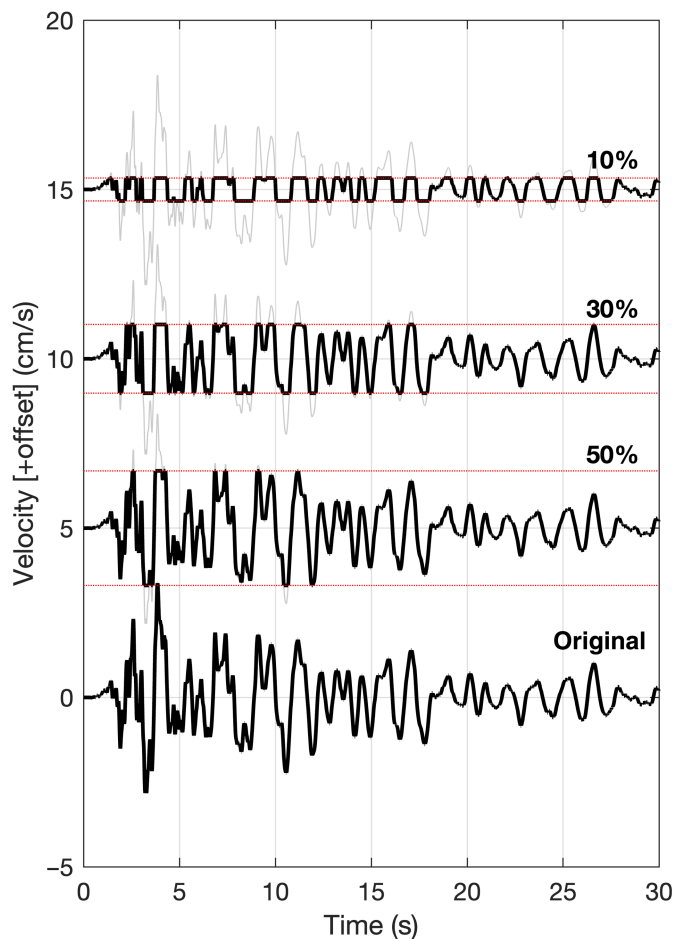


Figure 1. Example of artificially clipped European Strong-Motion (ESM)-database waveform (M 5.7 26 September 1997 00:33:12, horizontal record number E3002 at 32 km). From bottom to top: original waveform, 50%, 30%, and 10 % clip levels, with respect to peak ground velocity (PGV) (black). The original waveform is shown in light gray for reference throughout. The color version of this figure is available only in the electronic edition.

instruments were short period, single-component vertical seismometers, often coupled with 12-bit dataloggers (recording a maximum of 4096 discrete levels of motion). Datalogger sample rates were limited by available telemetry and storage capacity, meaning that recording 40, or even 20 samples per second was typical (resulting in Nyquist frequencies of just 20 or 10 Hz, respectively). In contrast, modern seismological networks record and broadcast continuous high-sample-rate (typically ≥ 100 samples per second) broadband data, from periods as low as hundreds of seconds to frequencies up to hundreds of Hertz. Recorded on triaxial sensors (sometimes at multiple depths) with 24-bit dataloggers, modern seismic installations provide high-fidelity records over a wide range of amplitudes, with over 16M discrete levels of motion.

A significant limitation of older seismic records was clipping, which typically manifests in digital time histories as a hard

upper (positive) and lower (negative) bound on recorded motions (Fig. 1). Clipping can be considered primarily not only as an indirect consequence of low-resolution digitization but also of hardware limits to strong motion. In the latter case, this may introduce “soft-clipping” (i.e., the output varies in some way with the input, but the relationship is no longer linear). In the past, owing to the limited resolution of 12-bit dataloggers, operators were forced to target a limited range of amplitudes and, consequently, earthquake scenarios. Otherwise, weak signals would be recorded with too few quantization levels (resulting in a “staircase-like” seismogram), or strong signals would saturate the datalogger (resulting in clipping). This balance was achieved through careful selection of instrument gain: in northern Europe, the target was typically to record weak ground motions [whether local microseismicity, regional seismicity (typically, $M < 4$), or teleseismic signals]. Therefore, instrument gain would be increased to ensure high-fidelity records covering the target range of weak motion, at the expense of reduced full-scale range (maximum recordable motion).

Even in the case of modern instrumentation with high-bit dataloggers, the full-scale range is limited by hardware. For instance, the Streckeisen STS-2, a stalwart of modern broadband seismology, has a full-scale range of ± 13 mm/s (up to 20 Hz) and will therefore clip during strong ground motion. For context, this means that median motions on rock from an M_w 5 at distances less than approximately 15 km, or M_w 6 at distances less than ~ 70 km (Boore *et al.*, 2014), are likely to clip on this sensor. Modern approaches to this problem include collocating strong-motion accelerographs, which can record accelerations up to $\pm 4g$ in the case of the Kinematics episensor. Nevertheless, this involves added expense, and as a result signal clipping is observed in numerous events from both the distant and more recent past.

Data reconstruction

According to Zhang *et al.* (2016), interpolation-based approaches to reconstruct clipped time-history data (e.g., Galbraith and MacMinn, 1982; Karabulut and Bouchon, 2007) are only suitable for limited degrees of data loss (defined by up to a few consecutive clipped samples). Yang and Ben-Zion (2010) introduced an improved Kriging-based interpolation method that was able to effectively deal with one-to-two consecutively clipped samples, leading to negligible errors. However, they noted that for three or more consecutively clipped samples, errors increased: visual inspection of their figure 10 shows peak amplitude errors increasing to $\sim 10\%$ for three clipped samples, $\sim 50\%$ for four samples, and 80% for five samples. For six or more consecutively clipped samples, errors using the interpolation method were in excess of 200% . For three or more clipped samples, Yang and Ben-Zion (2010) presented a “similar waveform” method, relying on a donor unclipped waveform, which generally provided better results. However, high (>0.88) cross-correlation coefficients between

the donor signal and the clipped signal were required to produce acceptable errors, limiting its application.

Zhang *et al.* (2016) overcame these limitations, introducing a clipping restoration method, herein referred to as “Clipped Waveform Restoration” (CWR), which is based on the “Projection onto Convex Sets” (POCS) approach. The POCS framework is well established in the field of seismic exploration as an advanced signal-processing tool, among many other fields, such as image processing. Applications in seismic data were initially focused on interpolation (Abma and Kabir, 2006), although numerous other techniques, such as denoising (Wang *et al.*, 2016), have since emerged.

Through synthetic analyses, Zhang *et al.* (2016) determined that when the clipping is weak, with “clip threshold” defined as discarding samples above 70%–90% of the peak amplitude, the restoration error for their POCS-based CWR method was 0.7%–1.7% of the peak amplitude; when clipped down to 40%–60% peak ground velocity (PGV), the reconstruction error increased to 3%–7% and for severe clipping, at 10%–30% PGV, errors were around 70%–90%. Cheng *et al.* (2023) recently implemented this approach to reconstruct 72 three-component clipped waveforms from the 2021 M_w 7.3 Madoi, China, earthquake. This allowed them to obtain more accurate S-wave phase-time arrivals to support their analyses of super-shear rupture. The drawbacks of Zhang *et al.*’s approach are that it is entirely deterministic, without consideration of uncertainty (beyond the benchmarking approximation of error based on clipping level), and it assumes that the reconstructed time history is stationary, such that its spectral characterization (amplitude and phase) does not vary with time.

Although modern machine learning (ML) approaches show promise for “bad-data” applications (e.g., Chen *et al.*, 2023), our focus here lies in extending and improving the established signal-processing POCS-based framework, as introduced by Zhang *et al.* (2016) for earthquake waveforms. Both traditional and ML approaches, which each have their own strengths, can then be used to appropriately account for epistemic uncertainty. We therefore build on the CWR method, through a time-variant implementation of the POCS approach, termed $CWR_{\Delta t}$. This allows a nondeterministic, nonstationary solution (with time-variant spectral properties) and associated temporal uncertainty to be established. The solution is subsequently propagated to alternative representations of the data, such as Fourier and damped response spectra, including uncertainties of these measures themselves. The definition of uncertainty thereby facilitates the use of clipped waveform data, previously deemed unusable, in a wide range of applications, from seismology to engineering.

We benchmark the newly developed $CWR_{\Delta t}$ method against ground-motion records from the European Engineering Strong-Motion (ESM) database (Luzi *et al.*, 2020), which we synthetically clip at increasing severity. We then show example application of the approach to the clipped

records of the 1992 M_L 5.8 Roermond earthquake, including at three distant sites in France, where collocated clipped (high-gain) and unclipped (low-gain) records exist. This provides an ideal real-world test of the $CWR_{\Delta t}$ approach, which is finally applied to data from the Netherlands, Belgium, and Germany to provide reconstructed time histories, and corresponding damped response spectra, with associated uncertainties, for scientific or engineering applications.

CWR

The CWR method (Zhang *et al.*, 2016) is based on Projection onto Convex Sets, an advanced signal processing technique common in image processing (e.g., Paek *et al.*, 1998) and with applications in exploration seismology (e.g., Abma and Kabir, 2006). Our implementation of POCS takes advantage of the fact that the Fourier amplitudes of clipped ground-motion records are similar to the underlying (unclipped/true) motion, albeit with contamination, or spectral leakage, associated with the missing data. The CWR method takes an iterative approach, starting with isolating the largest Fourier amplitudes on the basis that these are not, or are at least less severely, impacted by the spectral leakage associated with clipping. The underlying signal in these dominant harmonic components are reconstructed through the inverse Fourier transform and, where their time-history amplitudes exceed the full-band clipped signal, are retained in place of the original. The approach is then repeated, iteratively decreasing the threshold over which we isolate Fourier amplitudes of the partially restored record, thereby reconstructing the time history in a stepwise fashion.

Our $CWR_{\Delta t}$ approach builds on the work of Zhang *et al.* (2016) and leads to (1) reduced bias and record-to-record bias variability; (2) consideration of nonstationarity in the earthquake record; and (3) record- and time-specific estimates of restoration uncertainty. This is achieved primarily through:

1. splitting the full time history into multiple overlapping time segments, which have fixed duration and varying offset. The segment duration is defined by the record’s dominant frequency and apparent degree of clipping. Each clipped sample is then reconstructed multiple times from multiple segments, using different parts of the surrounding signal. This allows us to build a statistical model of the uncertainty in the reconstruction that is both waveform- and time dependent; and
2. introducing an optional frequency-content constraint on the reconstructed time history that imposes that the peak of the time derivative of the clipped waveform is retained, or at least not exceeded (within a defined tolerance), in the reconstructed waveform. For instance, in a velocity time history, we force the reconstructed velocity time history to have a peak acceleration that does not exceed, by more than a defined percentage, that in the (band-pass filtered) clipped waveform.

Our time-segmented approach is the key to addressing signal nonstationarity because it allows the spectral properties used for reconstruction to (at least partially) adapt over the duration of the time history, directly addressing the limitations of stationary methods that apply a single spectral constraint to the entire record. The multisegment approach also generates an ensemble of unique reconstruction pathways, and the resulting variance across these realizations provides a robust, data-driven measure of the temporal modeling uncertainty. The second constraint is related to our observations, during development of the $CWR_{\Delta t}$ approach for earthquake records, in which peak accelerations are typically unaffected by even moderate or strong clipping of velocity time histories. This constraint has a clear physical basis because the peak derivative (i.e., most rapid change in amplitude) of an earthquake time history usually occurs away from its peaks (amplitudes at the peaks themselves tend to roll over). This obviously has limits, because as clipping becomes more severe it is not only the peaks of the waveform that are truncated.

Clipping detection

The detection and labeling of clipped samples is an important initial step for the CWR methods. Clipping is identified in the record and flagged in a corresponding time history [$c(t)$, with 1 indicating clipped or potentially clipped samples and 0 for unclipped]. The detection of clipping is undertaken as a preprocessing step to the $CWR_{\Delta t}$ algorithm. Sophisticated methods for automatic clipping detection exist (e.g., Wang and Zhang, 2020) and may be necessary for complex cases, such as for soft-clipping in modern force-feedback systems. In fact, any clipping detection algorithm can easily be implemented alongside or instead of the detection approaches presented herein, which were found to be sufficient for the predominant “hard-clipping” observed in the low-resolution digitization of Roermond data.

In our approach for hard-clipped data (digitizer-resolution limited) the clip level is determined through two complementary methods based on: (1) peak amplitudes and (2) gradient detection. We define a record-specific amplitude histogram (equivalently a scaled PDF, Fig. 2b) of signal amplitudes. This is determined from the derivative of the cumulative distribution function of sorted time-history amplitudes, plotted against the fraction of peak amplitude. A time-history’s amplitude distribution is approximately zero centered in the absence of direct current (DC) shift or drift. Unclipped signals present a curve with a single broad peak in the histogram (i.e., only the central portion of Fig. 2b).

On the other hand, for clipped signals, two or three peaks are present (Fig. 2): one broad peak centered at zero (for demeaned time histories) and one or two peaks at the edges of the histogram, defining the upper and/or lower clip levels. For perfectly “hard-clipped” signals, we would observe zero-width spikes at the edges of the histogram (e.g., at 2047 counts and -2048 or -2047 counts, depending on digitization arithmetic, for

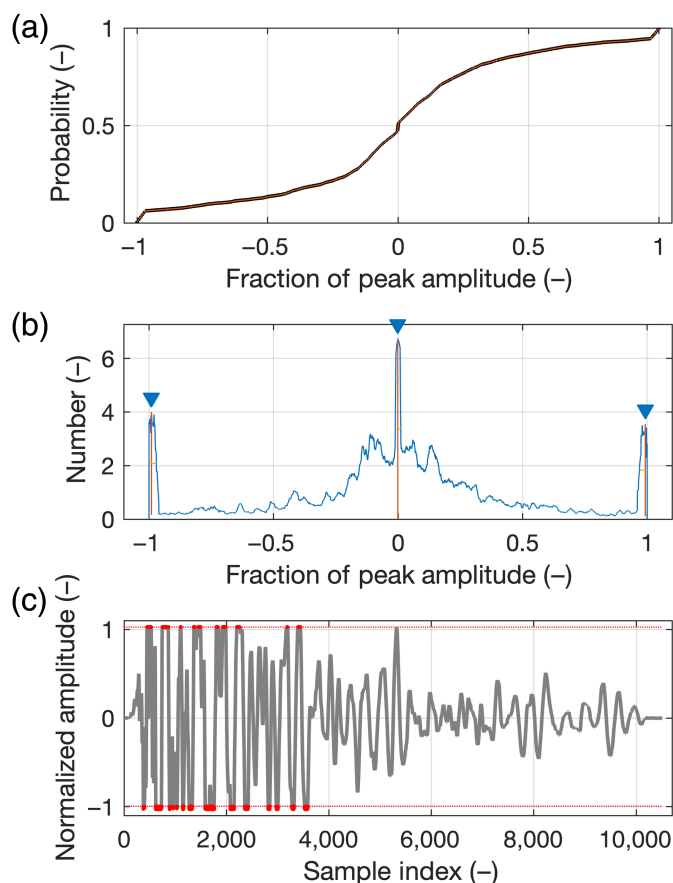


Figure 2. Example of clip detection using the data as in Figure 1 (soft-) clipped at $30\% \text{ of } PGV \pm 3 \text{ counts}$ (uniform distribution), normalized to ± 1 . (a) Cumulative distribution function (CDF) of amplitudes, with respect to PGV. (b) Histogram of signal amplitudes (derivative of CDF). (c) Clipped signal, with clipped samples (red) and upper/lower clip-level highlighted. The color version of this figure is available only in the electronic edition.

a 12-bit datalogger). However, in practice we frequently note a degree of “soft-clipping” in real data, resulting in nonzero standard deviation of the upper/lower clip amplitude distributions (e.g., the upper clip level may be observed at 2047 ± 3 counts). This is replicated in Figure 2 by clipping at discrete levels in each polarity: that is, $30\% \text{ of } PGV \pm 3 \text{ counts}$. To account for this minor “soft-clipping,” the clip levels in our detection algorithm are set at the median clip level detected, minus (for the upper clip level) the “half-width at half-height” value of the distribution of clipped amplitudes and vice versa for the lower clip level (i.e., placed at the zero-side edge of the clipped-amplitude distributions). To avoid detection artefacts (false peaks), we define a minimum peak-to-peak x -scale separation [0.6, in which the x -axis (amplitude) is normalized to ± 1 , as in Fig. 2] and a minimum peak-to-peak amplitude threshold (prominence). Minimum peak prominence is initially set at half the histogram maximum, then iterating with a 50% reduction (down to a defined minimum) until three peaks are detected. If more than three peaks

are detected, the iteration is repeated in the opposite direction (increasing peak prominence threshold) with a smaller interval (25%) until exactly three peaks are detected (the main distribution and the two clip levels). If the solution does not find three peaks, the waveform is reported as not being clipped. This may therefore miss clipped data that occur on one polarity only; these instances will nevertheless be picked up by the subsequent gradient-based clip detection algorithm. Given single-sided clipping, the amplitude-based algorithm can easily be modified to detect two dominant peaks in the histogram. Alternatively, as previously noted, any appropriate detection can be used to assign Boolean labels of clipped/not clipped to each time sample, including visual detection, or more sophisticated ML-based approaches.

The second clip-detection algorithm utilizes the fact that clipped samples follow a zero- (or near-zero) slope in the time history. This was implemented due to the difficulty in assessing clipping from amplitude distributions alone in weakly clipped examples, and because it was noted that in rare cases clipped samples were apparent for short duration within otherwise unclipped amplitude levels (i.e., we noted a systematic temporal variability of clip level in some old records). Although we are not aware of the physical mechanism for this latter case, it may indicate a more significant instrument malfunction. Nevertheless, these portions of data were clearly unusable and require restoration in the same way as the clipped data at the extrema of the waveform amplitude. In practice, we define potential clipping initially where the gradient is less than 1% of the observed resolution (quantization level, or counts) in the record (i.e., zero gradient, but allowing for small round-off errors). We subsequently remove the clip flag where the record shows low amplitude (<5% of the peak amplitude) and range (50 counts) across the ± 10 samples around the clip because this indicates low-amplitude signal quantization, rather than clipping. The final clip flag is defined based on the identification in either (1) amplitude or (2) gradient-based detection. It is important to note that the clip detection (particularly the gradient detection) should be applied to the raw digitizer output (in integer counts). Subsequent postprocessing, such as filtering or instrument response removal, masks the effect of clipping and makes detection difficult.

Finally, we implement a degree of conservatism when applying this approach, whereby amplitudes lying within a user-defined threshold (we use 15% for the Roermond data based on visual inspection) of the defined clip limit are also considered potentially clipped and treated no differently to clipped samples in the restoration. This ensures that only data samples with very low to zero likelihood of clipping remain untouched in the reconstruction algorithms.

Nonstationary clipped waveform restoration

The CWR _{Δt} approach, summarized by the flowchart shown in the supplemental material, available to this article, is detailed in

the following. Once clipped samples have been detected, we apply the reconstruction to N_p primary $\times N_s$ subsegments of the full time history, with the (maximum) segment duration (within each record) defined by

$$\Delta t_p = \frac{c_l}{f_c}, \quad (1)$$

c_l is a constant and f_c is the Fourier amplitude spectrum corner frequency, which can either be directly determined from the record or estimated. In our seismological application, we can estimate f_c from the earthquake magnitude and stress drop:

$$f_c = 0.4906\beta \left(\frac{\Delta\sigma}{10^{1.5M_w+9.05}} \right)^{\frac{1}{3}}, \quad (2)$$

(Boore, 2003), with $\beta = 3500$ m/s the shear velocity at the source, $\Delta\sigma = 10$ MPa the stress parameter, and M_w the moment magnitude. β and $\Delta\sigma$ can be assigned region-specific values, if available, but we found limited sensitivity to these parameters in testing. We find that linking the segment length to the characteristic frequency of the signal, through f_c , results in improved reconstructions. This is likely due to the fact that the segment length is then linked to the dominant number of cycles in the time series.

Each of the primary time segments considered for reconstruction is overlapped on every iteration through the full time history by a defined proportion of the window length (e.g., using 50% will overlap successive segment windows with half of the previous, see Fig. 3, top). The N_p overlapping time segments of the full time history are used to provide an iterative update in the time domain: the $(p + 1)$ th segment uses part of the (partially) reconstructed prior p th segment as an input. The clipping flags are not updated in between these overlapping iterations. This results in iterative updates to the amplitudes of overlapping portions of segments.

Within the overlap of the p th time segment (O_p), N_s subsegments (offset by defined intervals, O_s) up to the full overlap of the p th time segment are used to provide alternative realizations of the reconstruction (see Fig. 3, bottom and middle). We use this to build a statistical representation of the modeling uncertainty in each of the primary segments. For example, if the P segments are defined with 50% overlap ($O_p = 0.5$), a 5% offset-overlap ($O_s = 0.05$) results in $N_s = 11$ steps (and 11 unique reconstruction realizations) between iteration p and $p + 1$. Alternatively, this can be thought of as $S = 11$ unique windowing strategies with increasing offset, from zero to the full overlap (see Fig. 3 for an example with $N_s = 3$). Each of the S realizations are retained independently as unique estimates of the full time-history restoration. The mean and sample standard deviation of the S time-aligned reconstructions are then used to reconstruct a single “best-estimate” time history with defined uncertainty.

The CWR _{Δt} approach therefore involves the stepwise reconstruction of up to $N_p \times N_s$ segments $[x_{ps}(\tau)]$ of the full

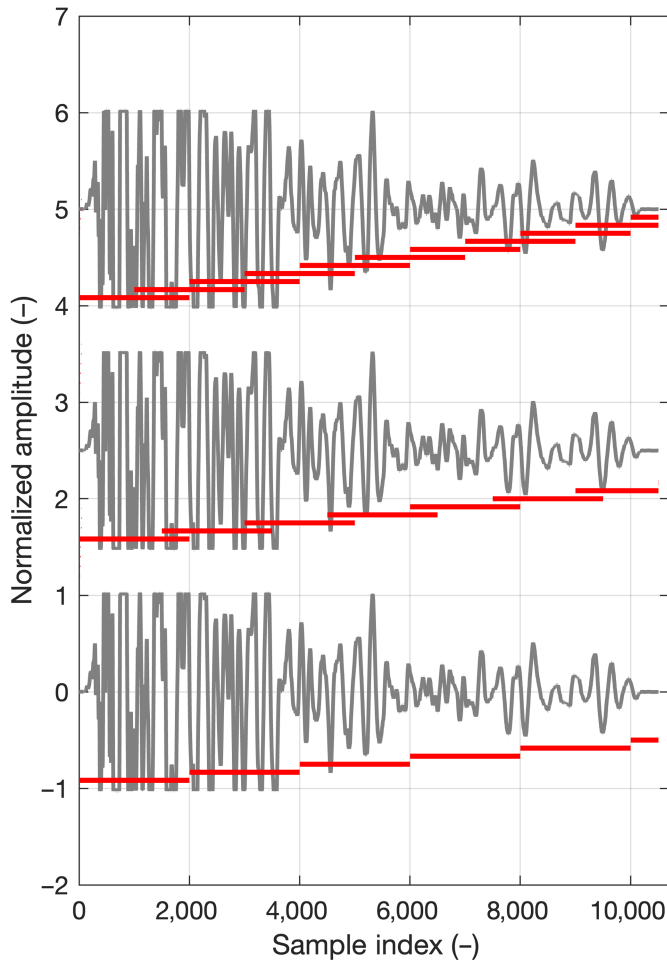


Figure 3. Simplified example of windowing for $CWR_{\Delta t}$. Here (primary, or maximum) overlap is $O_P = 0.5$ (50%), with window lengths of 2000 samples (top). The red lines indicate the location of successive windows in each analysis. Secondary offset in this example is defined as $O_S = 0.25$, such that a further two windowing schemes are produced, one with zero overlap (bottom) and one with 25% overlap (middle). Reducing O_S would result in additional unique windowing schemes (e.g., $O_S = 0.05$ would result in 11). The color version of this figure is available only in the electronic edition.

time history, $y(t)$. Each segment $x_{ps}(\tau)$, with τ the time within a given segment, has a duration of Δt_p (equation 1). The s th sub-segment of the p th time-segment iteration (defining the specific time series window for reconstruction) is defined as

$$x_{ps}(\tau) = \begin{cases} y(t - t_{\min_{ps}}) & t > t_{\min_{ps}}, \quad \tau < \Delta t_p \\ 0 & \text{otherwise} \end{cases}, \quad (3)$$

in which the start time of each segment $x_{ps}(\tau)$ is defined by

$$t_{\min_{ps}} = (p-1)\{O_P + [(s-1)O_S]\}\Delta t_p \quad |(s-1)O_S| \leq O_P. \quad (4)$$

For each segment, $x_{ps}(\tau)$ (recall that this must be in integer units that are clipped—i.e., digitizer counts, before any processing), its FAS, $X_{ps}(f)$, is determined. The inverse FFT (iFFT) is then computed over N_a iterations of a threshold ($0 < \alpha_n < 1$) of the peak of the FAS. The n th iFFT comprising $m_\tau \in \mathbb{Z}^+ \leq M_\tau$ samples, is given by

$$x_{psn}(\tau_m) = \frac{1}{M_\tau} \sum_{k=0}^{M_\tau-1} \begin{cases} X_{psn}(f_k) & |X_{psn}(f_k)| > \alpha_n \times \max(|X_{psn}(f_k)|) \\ 0 & \text{otherwise} \end{cases} e^{i\frac{2\pi}{M_\tau}km}. \quad (5)$$

In practice, $x_{psn}(\tau)$ is the result of an inverse FFT on the values in the spectrum $X_{ps}(f)$ exceeding the threshold value α_n of the peak of the spectrum, with other terms zeroed. The reconstructed time-history segment is then determined, $x_{ps}(\tau)$, by iteratively retaining the maxima of the (partially) reconstructed signal $x_{ps}(\tau)$, and the iFFT result, $x_{psn}(\tau)$:

$$x_{ps}(\tau) = \begin{cases} x_{psn}(\tau) & |x_{psn}(\tau)| > |x_{ps}(\tau)| \\ x_{ps}(\tau) & \text{otherwise} \end{cases}. \quad (6)$$

In an optional step, the reconstructed trial time-history segment $x_{psn}(\tau)$ is checked to ensure its first time derivative maxima does not exceed (within a user-defined tolerance, e.g., 5%) that of the clipped waveform. If it does, $X_{psn}(f)$ is low-pass filtered (by convolution with a complex frequency-domain fourth-order Butterworth filter) using an iteratively reduced cutoff frequency, f^{lp} , (down to a minimum of the square root of the Nyquist) until compliance. Over the N iterations, α is linearly reduced toward zero until the full spectrum $X_{psn}(f)$ is used to obtain $x_{psn}(\tau)$, but recalling that the time history $x_{ps}(\tau)$ has been iteratively updated (equation 6) to produce $X_{psn}(f)$. It is important to note that the full time history is not filtered (at f^{lp}), only the partial reconstructions x_{psn} based on the iFFT of spectral amplitudes above the threshold α_n . Filtering of the full reconstructed time history to account for unwanted spectral content, such as noise in the unclipped samples, may be deemed necessary at a later stage in the data processing workflow. That is, however, not dealt with here, with the focus rather on the reconstruction of high-fidelity ground-motion records, irrespective of whether those compose signal or noise.

After iteration over S , and alignment with respect to time, t , the mean amplitude of the p th segment is determined by averaging over the S iterations:

$$x_p(t) = \frac{1}{S} \sum_{s=1}^S x_{ps}(t). \quad (7)$$

Record- and time-dependent modeling uncertainty is defined by

$$\sigma^2[x_p](t) = \frac{1}{S-1} \sum_{s=1}^S [x_p(t) - x_{ps}(t)]^2. \quad (8)$$

Note that as the P time segments are partially overlapped, parts of $x_p(t)$ and $\sigma^2[x_p](t)$ will update on subsequent iteration, such that overlapping sections of $x_p(t)$ and $\sigma^2[x_p](t)$ are identical. Finally, $x(t)$ and $\sigma^2[x](t)$, the full reconstructed time history and its uncertainty, are obtained by joining (accounting for appropriate overlap) segments $p = 1$ to N_p .

Benchmark testing

To test the CWR_{Δt} method, we make use of the database used in the development of the Akkar *et al.* (2014) “Ground-Motion Models for Europe and the Middle-East,” which is a subset of the ESM database (Luzi *et al.*, 2020). A total of 452 records from sites within 150 km of moderate or strong earthquakes with magnitudes between 4.5 and 6.5 are used in our analyses. The records were used as processed and individually low-frequency filtered by Akkar *et al.* (2014). Clipping is most likely to occur on seismometers because accelerometers typically have a full-scale range of $\pm 2g$ or even $\pm 4g$, which is sufficient for even the strongest earthquake ground motions. Seismometers, on the other hand, have more modest clip levels. For this test, we therefore imagine the database had been recorded on seismometers, converting all records to units of velocity through integration of the acceleration time histories. A low-pass sixth-order Butterworth filter with corner at 0.1 Hz is applied prior to integration. These velocity waveforms, $y'(t)$, are then artificially clipped to produce the input time history $y(t)$, with increasingly severe clip levels (90%–10%, inclusive, in steps of 10%), defined as a percentage of the PGV of the original time history (see Fig. 1). We then test the restoration using an increasingly large number of time segments (P , with segment length defined by $c_l = 1000, 20, 10, 7$, and 5 in equation 1). $c_l = 1000$ is large enough that it corresponds to using the full waveform and is most similar (but not entirely equivalent) to the original approach of Zhang *et al.* (2016). $c_l = 5$ for a record of an M_w 4.5, 5.5, and 6.5 event would correspond to 2.5, 7.9, and 25 s segment lengths, respectively, whereas intermediate values of c_l lead to proportionally longer segment lengths. In this benchmark test, for the sake of computational efficiency, we do not iterate over the secondary windowing. This results in a single deterministic (albeit potentially nonstationary) waveform, $x(t)$, without amplitude uncertainty.

The reconstruction performance (with respect to the original, unclipped velocity waveform) is measured in several ways. The difference between the reconstructed and original time histories is the simplest method. We measure misfit only for samples identified as clipped or potentially clipped [i.e., clip flag, $c(t) = 1$] to avoid the bias of zero-misfit unclipped samples.

The record misfit is defined as the average of the modulus of difference between the unclipped and reconstructed signal:

$$M_t = \frac{1}{\sum c(t)} \sum c(t) |y'(t) - x(t)|. \quad (9)$$

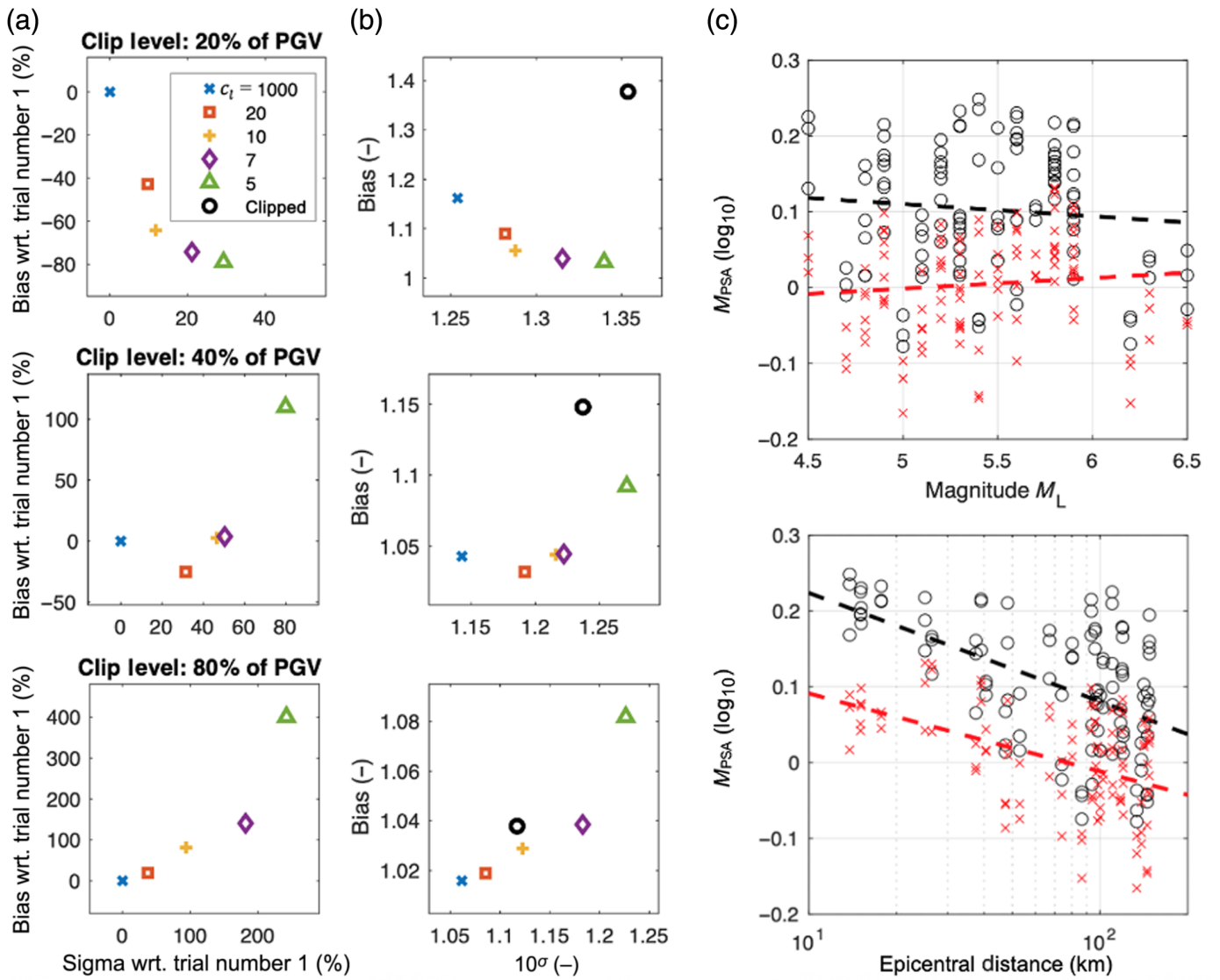
For the purposes of engineering seismology, the difference between damped response spectral ordinates may be considered a more useful metric. We therefore measure bias and standard deviation of the reconstructed versus unclipped 5% damped pseudospectral response spectra (PSA), $Y(T)$ and $Y'(T)$, respectively, over 23 periods from peak ground acceleration (PGA) (0.01 s) to 1 s. Specifically, we use periods 0.01, 0.02, 0.025, 0.03, 0.04, 0.05, 0.075, 0.1, 0.12, 0.125, 0.15, 0.17, 0.175, 0.2, 0.25, 0.3, 0.4, 0.5, 0.6, 0.7, 0.75, 0.85, and 1.0 s. The period specific misfit for each record is then defined as

$$M_{\text{PSA}}(T) = \log_{10} Y'(T) - \log_{10} Y(T). \quad (10)$$

The average of the misfit values, $M_{\text{PSA}}(T)$, over all periods is denoted as M_{PSA} , and the average of this over all records as $\langle M_{\text{PSA}} \rangle$ with standard deviation $\sigma_{M_{\text{PSA}}}$. Similarly, we determine averages using the modulus, $|M_{\text{PSA}}(T)|$, denoted $|M_{\text{PSA}}|$ and $\langle |M_{\text{PSA}}| \rangle$. An example of M_{PSA} for clip values at a severe level 20% of PGV is shown in Figure 4c. Misfit is independent of magnitude but appears to be distance dependent for both clipped and reconstructed time series, despite the clip level being constant and record specific. We are not aware of the reason for this distance dependence in clip restoration misfit, although it may be related to the increasing duration and characteristic envelope shape of more distant signals. Nevertheless, the restored waveforms exhibit minimal overall bias, unlike the clipped waveforms.

Figure 4 shows how the measure of absolute misfit ($|M_{\text{PSA}}|$) is used to determine optimal values of the window length parameter c_l . We observe that $c_l = 10$ represents an optimal window segmentation (i.e., 5 and 15.8 s for M_w 4.5 and 5.5, respectively), with low bias and record-to-record bias variability, termed sigma, for modest- to strongly clipped data (i.e., clipped at 40% of PGV or below). This is a trade-off in which, for severe clipping, $c_l = 10$ is shown to significantly reduce the bias $10^{\langle |M_{\text{PSA}}| \rangle}$ compared to $c_l = 1000$ (Fig. 4b, top). This comes at the cost of a very minor increase in sigma at moderate or low clip levels (Fig. 4b, bottom). At severe clip levels of 20% PGV, the unrepaired clipped signal results in an average modulus bias in the PSA of 38%, with significant record-to-record variability (standard deviation of the record-to-record variability in misfit equivalent to 35%). Using $c_l = 10$, this bias is reduced to 5.5% with lower record-to-record variability. For reference, Zhang *et al.* (2016) reported that for similarly severe clipping (10%–30% PGV), they observed errors of around 70%–90%.

For weakly clipped data (e.g., clipped at 80% of PGV), typically resulting in only the highest amplitude peaks being

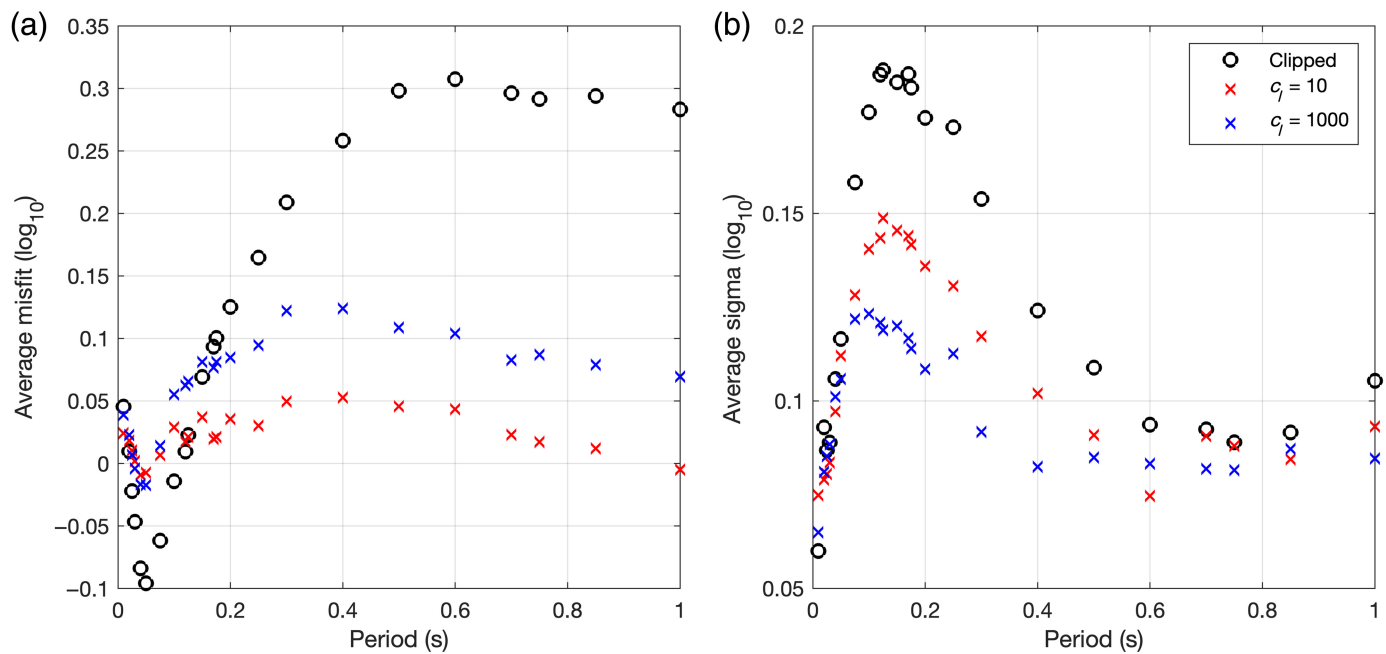


saturated, time segmentation slightly increases the average bias compared to reconstructions without. CWR without time segmentation ($c_l = 1000$) is technically the best approach here in terms of minimizing reconstruction bias. However, the biases introduced by time segmentation are very low (and in all cases with $c_l \geq 10$ still provide an improvement on the clipped signal). For example, for the weakly clipped signals at 80% PGV and using $c_l = 10$, we observe average absolute bias in spectral ordinates, $\langle |M_{PSA}| \rangle$, of 2.9%, a reduction of 0.8% compared to that obtained from the clipped records. On the other hand, we observe a slightly smaller $\langle |M_{PSA}| \rangle$ of 1.6% for $c_l = 1000$ in this case. This suggests that although a targeted approach may be optimal, simply using $c_l = 10$ will result in satisfactory reconstructions for all scenarios and does not involve having to estimate the unknown clip level a priori.

Figure 5 shows the spectral misfit and its standard deviation, in terms of $M_{PSA}(T)$, averaged over all records, as a function of period for: records clipped at 20% of PGV; restoration without time segmentation ($c_l = 1000$); and using the time-dependent

Figure 4. Bias factor, $10^{|M_{PSA}|}$, (of clipped and reconstructed time histories with respect to unclipped originals) and its standard deviation (shown in terms of 10^{σ}). (a) Bias and sigma presented as percentage change with respect to (wrt.) trial 1 with $c_l = 1000$ (i.e., without time segmentation). (b) Bias and sigma as absolute values. Note that in both panels (a and b), axis limits (both bias and sigma ranges) vary depending on clip level. (c) Record M_{PSA} plotted against local magnitude (a) and distance (c) for both clipped and restored records. PSA, pseudospectral response spectra. The color version of this figure is available only in the electronic edition.

approach, with $c_l = 10$. The most significant impact of clipping on PSA comes at the mid- to long periods, with up to a factor of 2 ($[M_{PSA}(T) \approx 0.3]$ reduction in PSA in this scenario. The reconstructed time histories reduce this bias significantly, with the optimal case $c_l = 10$ leading to misfits consistently below 10% [$M_{PSA}(T) < 0.05$] across all periods. We note that the trade-off between reduction in misfit and corresponding



increase in record-to-record variability (although small), as seen in Figure 5, is broadly period independent.

1992 Roermond Earthquake

We use the 1992 Roermond earthquake (M_L 5.8) as a real-world test case for the $CWR_{\Delta t}$ method. This normal dip-slip event struck in the early hours of the morning, at 01:20:02 UTC on 13 April. Its epicenter lay approximately 5 km southwest of the town of Roermond in the Netherlands (51.16°N, 5.95°E), close to the borders of Belgium and Germany in the extensional Roer Valley Graben. Measured at M_w 5.3–5.4 (Braunmiller *et al.*, 1994; Camelbeeck *et al.*, 1994), the earthquake still stands as the largest instrumentally recorded on mainland northwestern Europe (Lammers *et al.*, 2023), and therefore as a reference event for earthquake ground motion in this region. Precise focal depth is uncertain, with estimates between 14 and 25 km depending on the approach used (Dufumier *et al.* 1997, and references therein; Spetzler *et al.*, 2024). Despite the deep focus, macroseismic shaking intensity reached MSK VII (very strong) at the epicenter (Haak *et al.*, 1994), with reports of ground failure including liquefaction and sand boils (Nieuwenhuis, 1994), and landslides (Alkema *et al.*, 1994; Maurenbrecher *et al.*, 1994) in nearby Brunssummerheide.

The earthquake was well recorded at local and regional distances, mainly by national networks of the Netherlands, Belgium, and Germany. More distant records are also available across Europe and even globally due to the size of the event. The closest seismic station, located 33 km from the epicenter, was the Dutch site KRN (Camelbeeck *et al.*, 1994), yet data from this and several other sites was, until now, considered unusable due to clipping.

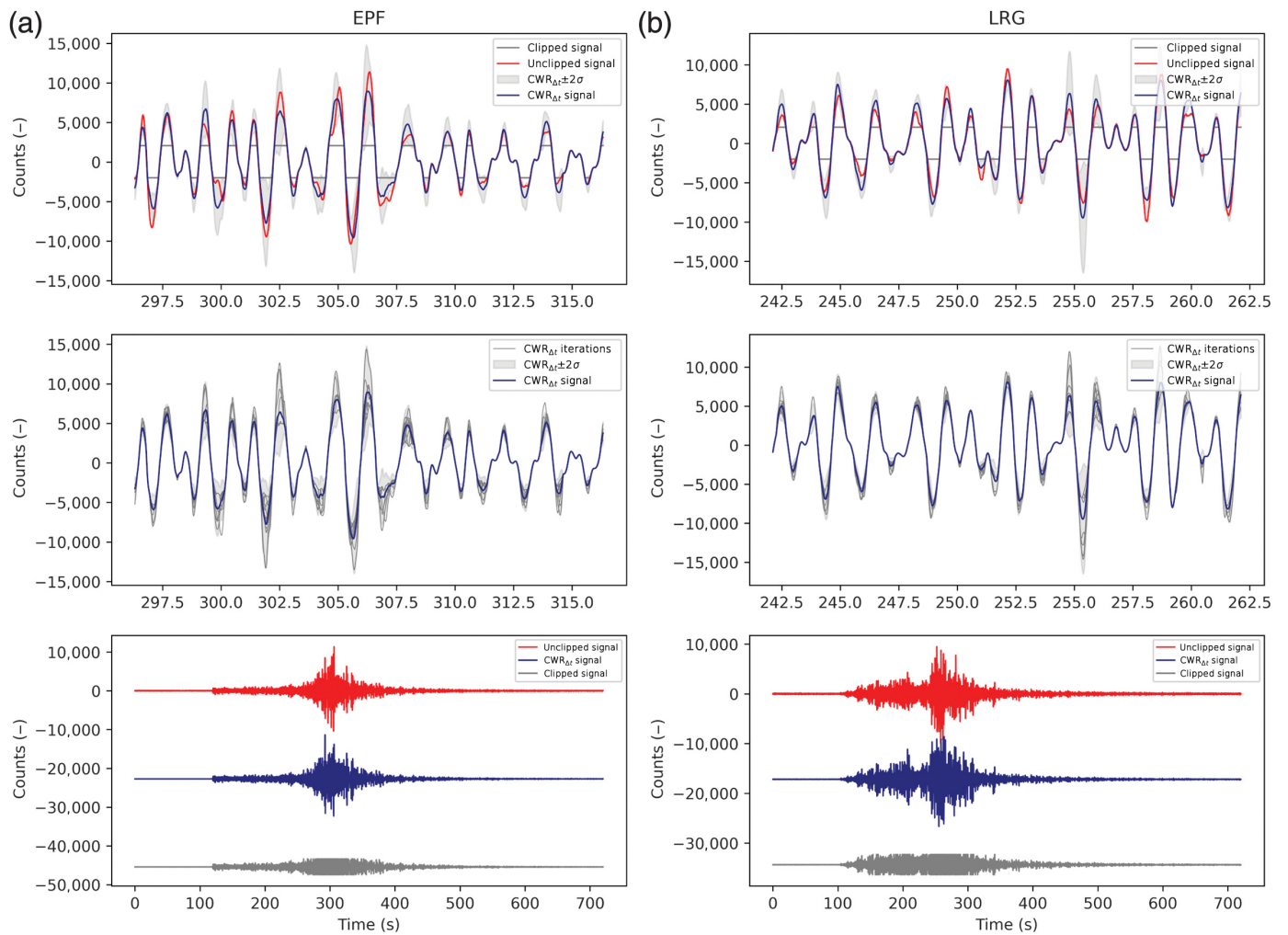
In early analysis of records, Gariel *et al.* (1994) reported that the closest Belgian operated station “which did not saturate”

Figure 5. Misfit [log(bias)] (a) and their standard deviations (b) at specific periods for the benchmark test with time histories severely clipped to 20% of the original PGV. $CWR_{\Delta t}$ reconstructions without time segmentation ($c_j = 1000$) and with $c_j = 10$ are shown along with the results directly from the clipped records. The color version of this figure is available only in the electronic edition.

(VIA) was located in Vianden, Luxembourg, located at an epicentral distance of 138 km. However, close inspection reveals, even in this case, the peak velocity on the horizontal component (i.e., the S wave) was slightly clipped. The closest entirely unclipped record available at this time is from the German station BUG in Bochum, Germany. This is an STS-2 broadband sensor located 98 km from the epicenter. This is a useful example highlighting that epicentral distance and ground motion alone do not dictate clipping: the calibration and digital-mechanical characteristics of the instrument are crucial. In fact, most records from (high-gain) seismic networks in the United Kingdom, France, and Switzerland were severely clipped, despite being significantly farther away (even at distances well in excess of 1000 km).

Real-World Benchmark

Building on the synthetic test applied to the strong-motion records of the ESM database, we utilize three records from the 1992 M_L 5.8 Roermond earthquake as a real-world benchmark for the $CWR_{\Delta t}$ approach. These records are from the French Laboratory for Detection and Geophysics (LDG) network where, despite the majority of records on the network’s short-period vertical seismometers clipping, collocated low-gain sensors were available at a few sites, three of which (EPF, LRG, and FLN) are not affected by clipping.

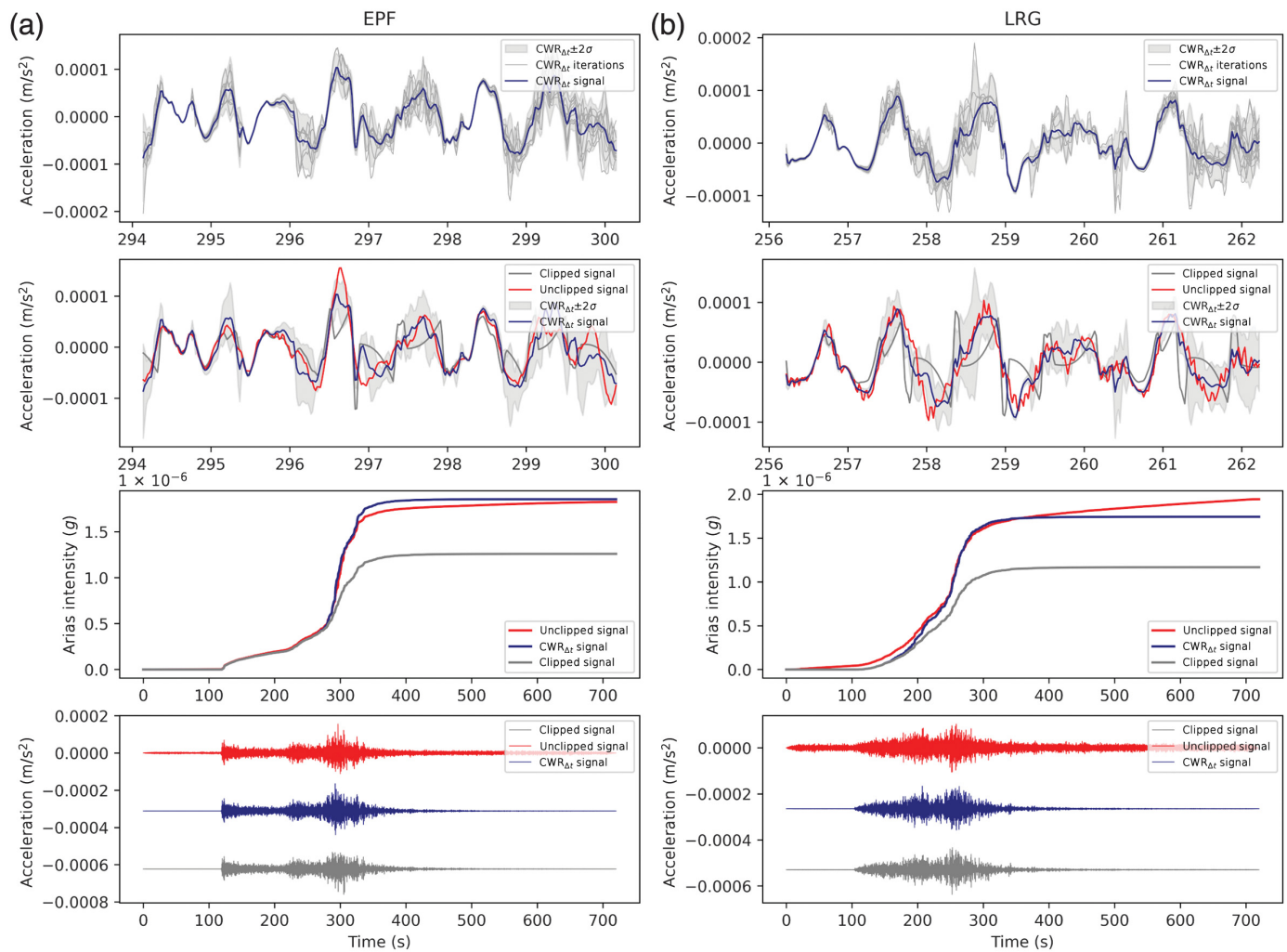


Unfortunately, the records are several hundred kilometers from the epicenter. However, the benefits of the real-world benchmark through collocated low-gain (unclipped) reference time history are considered to outweigh this limitation.

In implementation here, primary segment overlap of the CWR_{Δt} method is set to $O_p = 0.5$ (50%), with $O_s = 0.025$ (2.5%) secondary steps providing $S = 21$ reconstruction realizations to define the mean and variability of the reconstructed time history. In testing this was found to balance computation time with uncertainty quantification. Using a range of alternate values may be useful in exploring epistemic uncertainty in waveform reconstruction. Site EPF was located at 999 km from the epicenter. By comparing the unclipped and clipped signals, we determine that the high-gain record is severely clipped, with clip level at 18% of PGV. This corresponds to a maximum single-polarity loss of 9313 counts. The average amplitude error across all clipped points is 29.7%, equivalent to 40 counts on average across all clipped (or potentially clipped) samples in the record. This highlights that a significant amount of clipping occurs away from the peak of the signal. The reconstruction (Fig. 6a) shows good performance, despite the severe clip level. The restored signal shows an average time-history amplitude error of -12.7% (the

Figure 6. CWR_{Δt} reconstruction of sites (a) EPF and (b) LRG, clipped at 18% and 28% of PGV, respectively. The bottom figures indicate clipped, reconstructed, and unclipped reference signal. The middle figures indicate focus on PGV, showing individual CWR_{Δt} reconstructions (gray lines) used to determine mean and uncertainty. The top figures indicate middle, with clipped, unclipped, and CWR_{Δt} reconstructed waveform including ±2 standard deviation (95% confidence level).

negative implying amplitudes are, on average, slightly higher than the reference unclipped signal). This equates to an average clip error of 10 counts (reduced from 40), with the maximum absolute error reduced by 64% from 9313 to 3333 counts (note that the maximum error in the reconstructed signal was not at the peak velocity). The peak amplitude in the low-gain unclipped, gain adjusted, time history (i.e., our reference) was 11,382 counts, with the clipped signal limited to 2069 counts (including DC offset). The CWR_{Δt} reconstruction has a peak of 11,241 counts, which is within 1.2% of the true peak value of 11,382 counts, significantly below the reconstruction error of 70%–90% reported by Zhang *et al.* (2016) for signals clipped at this severity.



In terms of PGA, the clipped record led to a 12.1% loss in recorded amplitude, which was reduced to a loss of only 7.1% in the reconstructed waveform (Fig. 7a). The Husid plot (e.g., Bommer and Martínez-Pereira, 1999), representing the cumulative Arias intensity with time, shows the significant cumulative impact of clipping in the acceleration domain, even if the peak values are comparatively unaffected. The repaired signal almost perfectly matches the reference values. This highlights that although peak values are important, other characteristics of the waveform are successfully recovered through the $CWR_{\Delta t}$ process.

Site LRG (Fig. 6b) was located at 859 km from the epicenter, with the high-gain record clipped at 28% of PGV, slightly less than the previous record, but still severe. The restored signal shows an average error across all clipped samples of -20.7% , or -10 counts (compared to the clipped signal with 22.6% , or 39 counts), again indicating on average an overestimation, albeit fairly minor, of amplitudes in the reconstruction. The reconstruction (Fig. 6b) shows good performance, despite the severe clip level. The maximum absolute error decreased by 57% from 7929 to 3432 counts (again, the maximum error in the reconstructed signal was not at the peak velocity).

Figure 7. $CWR_{\Delta t}$ reconstruction of sites (a) EPF and (b) LRG, as Figure 6 but differentiated to acceleration. The bottom figures indicate velocity clipped, reconstructed, and unclipped reference signal (in acceleration). The lower middle figures indicate Husid plots showing cumulative Arias intensity. The upper middle figures indicate focus on peak ground acceleration (PGA), indicating reference unclipped, clipped, and $CWR_{\Delta t}$ reconstruction including ± 2 standard deviation (95% confidence). The top figures indicate as previous but showing only $CWR_{\Delta t}$ iterations along with the mean.

The peak amplitude in the low-gain unclipped, gain adjusted, time history (i.e., our reference) was 9927 counts, with the clipped signal limited to 2067 counts (including DC offset). The $CWR_{\Delta t}$ reconstruction has a peak of 9164 counts, which is within 7.7% of the true peak value and again significantly better than reported by Zhang *et al.* (2016) for signals clipped at this severity.

In terms of PGA at site LRG (Fig. 7b), the clipped record led to only a 1.7% gain in recorded amplitude with respect to the reference signal. This was actually reversed to a loss of 12.9% in the reconstructed waveform. The Husid plot shows a

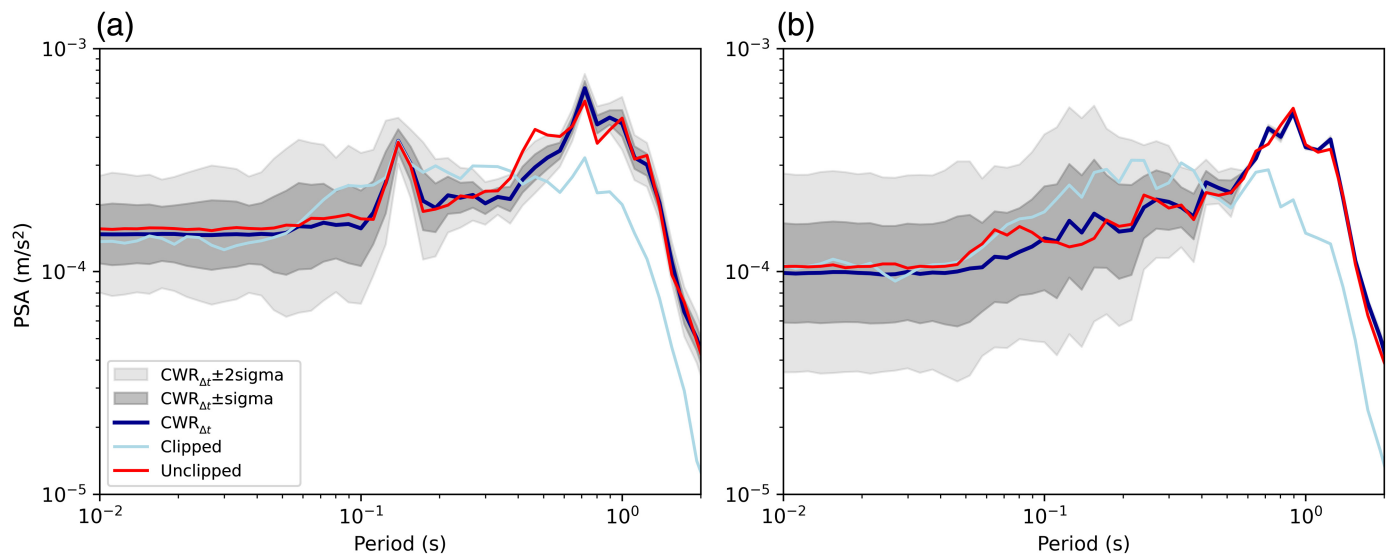


Figure 8. 5% damped pseudoresponse spectra for sites EPF (a) and LRG (b), showing the clipped spectra, the reference unclipped, and the $CWR_{\Delta t}$ repaired spectra along with ± 2 standard deviations (95% confidence limits). The color version of this figure is available only in the electronic edition.

significant improvement in the restored waveform with a close match to the reference. In fact, the reconstructed acceleration waveform is arguably a better representation than the “reference” in this case due to the fact that this low-gain “reference” record appears to have significant noise that leads to increasing Arias intensity with time after the passage of the significant shaking. This noise artefact is not present in the reconstructed signal. It is also noteworthy that although the peak acceleration of the clipped waveform closely matches the peak of the reference, the acceleration time history of the clipped signal itself is very erratic due to the artefacts introduced in differentiating the truncated velocity signal. Naturally, it would be difficult to isolate these effects without the reference acceleration, unlike for the native ground motion (velocity) of the sensor (where values are clearly visually truncated). It is clear, however, that the reconstructed acceleration traces (at both LDG sites where the reconstruction was successful) appear more visually consistent with the reference waveform and our expectations of a typical acceleration time history despite, in the latter case, a slight increase in the mismatch of peak acceleration amplitude.

A third site with collocated instrumentation, site FLN (included in the supplemental material) was located at 532 km from the epicenter. By comparing the unclipped and clipped signals, we observe that the high-gain record is clipped down to a level corresponding to just 7% of the true PGV. The result of this extreme level of signal loss is that the waveform was not adequately reconstructed, evident through visual inspection of the result.

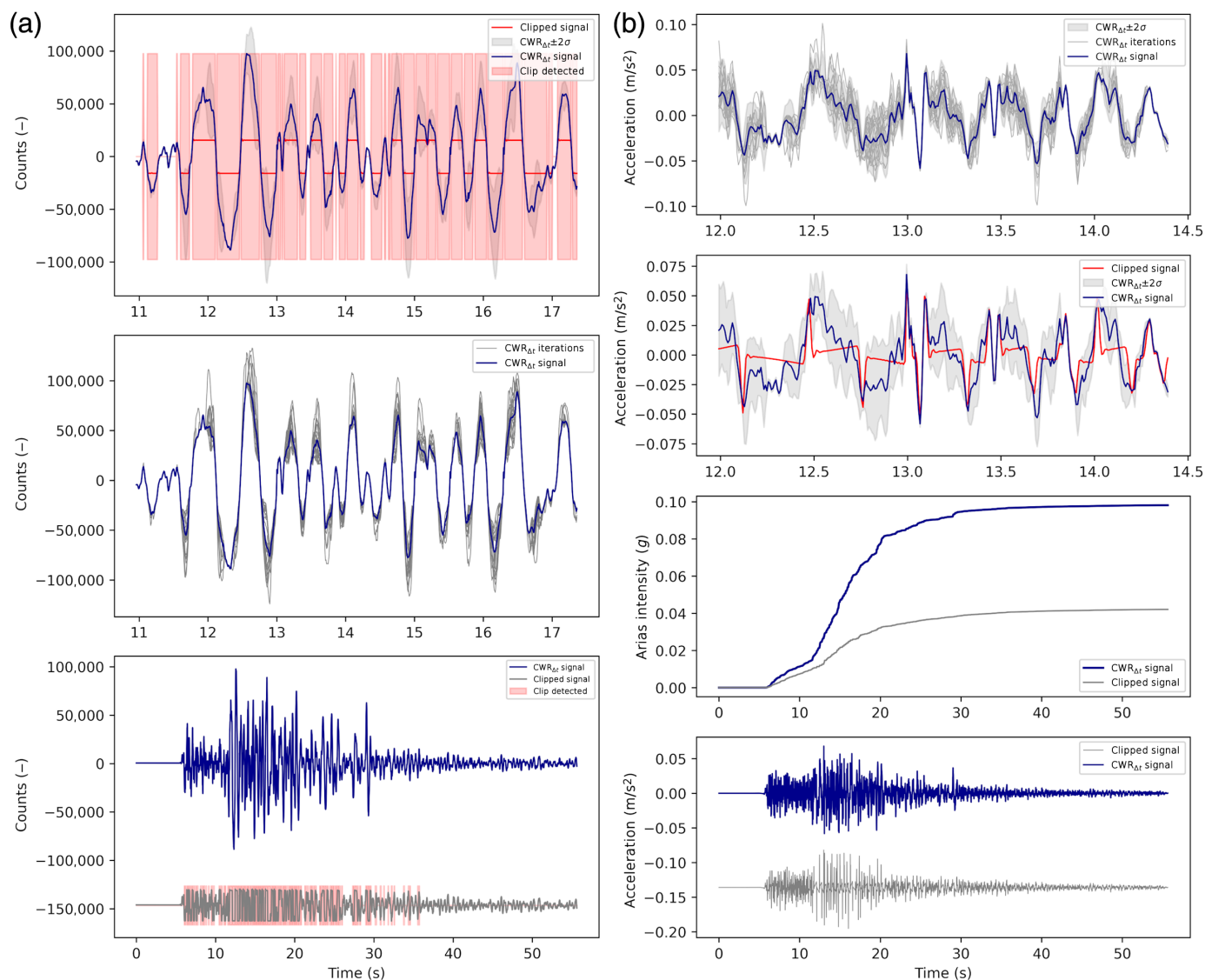
This real-world test highlights the strong performance of the $CWR_{\Delta t}$ method. First derivatives seem to be well retained even in the clipped record, despite the significant clipping. In fact, in the case of site LRG, the PGA was almost unaffected by the severe clipping (despite the resulting acceleration time history looking unusual). Performing restoration on the signal led to a slight increase in the mismatch of PGA with that of the low-gain signal, but it also resulted in a significant

improvement in other characteristics of the acceleration waveform, such as Arias intensity and coherence. In all cases, the errors in the reconstruction are certainly well within acceptable tolerances for use in engineering applications, as demonstrated through the comparison of PSA for the clipped, reference unclipped, and $CWR_{\Delta t}$ waveforms (Fig. 8). Clipped waveforms tend to lead to a reduction in long-period PSA and an overestimate of midperiod PSA. The ability to define the temporal uncertainty of the time-domain reconstruction, and how it propagates through to PSA (among others), means that we can, in theory, include such peak measures along with their uncertainty in model derivation, such as GMPEs.

Near-Source Roermond Records

We finally demonstrate the application of the $CWR_{\Delta t}$ approach on some records from the epicentral region of the 1992 M_L 5.8 Roermond earthquake. These data were recorded at distances from around 35 km, mainly by the Belgian seismic network. In this case, we have no reference waveform for comparison. However, we can rely on the success of the approach on the synthetic and real-world examples with “ground-truth” records. A significant additional problem in these records was identified in that clipping exhibited wrap-around, with clipped negative signals being recorded as clipped positive. This resulted in artefacts and spikes in the record, particularly when differentiated to acceleration.

The wrap-around effect likely results from digitizer overflow. In effect, an input that should read as “more negative” than the lower limit appears to roll over and show up as a



positive full-scale value. The $CWR_{\Delta t}$ method requires the clip polarity to be correct, meaning that this wrap-around artefact had an adverse impact on the reconstruction. We solved this by assuming that the polarity of clipped samples was only recorded by preceding unclipped samples. In a preprocessing step, clipped samples were therefore assigned polarity of the most recent unclipped sample.

Two examples are shown here: the first for the vertical record BE.BEBN at 46 km from the epicenter (Fig. 9), which is the closest Belgian record available. The record was, based on our reconstruction here, clipped to 17.2% of PGV. The second example is the record BE.CLA, recorded at a distance of 95 km (Fig. 10). This record was again heavily clipped (at 24.4% of PGV), with significant polarity reversal through wrap-around errors in the datalogger. Despite the severe level of clipping, for both cases waveforms are generated that appear realistic (and based on the synthetic and real-world testing) can be assumed to provide representative ground motions that are usable within other seismological and hazard products.

Figure 9. Record BE.BEBN (single Z component) at 46 km from the epicenter. (a) Native digitizer counts (proportional to velocity), as Figure 6. (b) Acceleration-based data, as Figure 7. Detected clipping is highlighted (left top and bottom panels). The color version of this figure is available only in the electronic edition.

Conclusions

We presented a POCS-based method for clipped waveform restoration, $CWR_{\Delta t}$, which builds on the approach of Zhang *et al.* (2016). The new algorithm accounts for signal nonstationarity and provides temporally variable reconstruction uncertainty. Reconstruction errors appear to be significantly lower than those reported by Zhang *et al.* (2016) at high levels of clipping (and comparable otherwise). For severe clipping (clip level at 20% of PGV), mean reconstruction error was 5.5%; for moderate clipping (50% of PGV) the mean error was 5.0%; and for minor clipping (80% of PGV) the mean error was 2.9%. For comparison, Zhang *et al.* (2016) reported errors of 70%–90%,

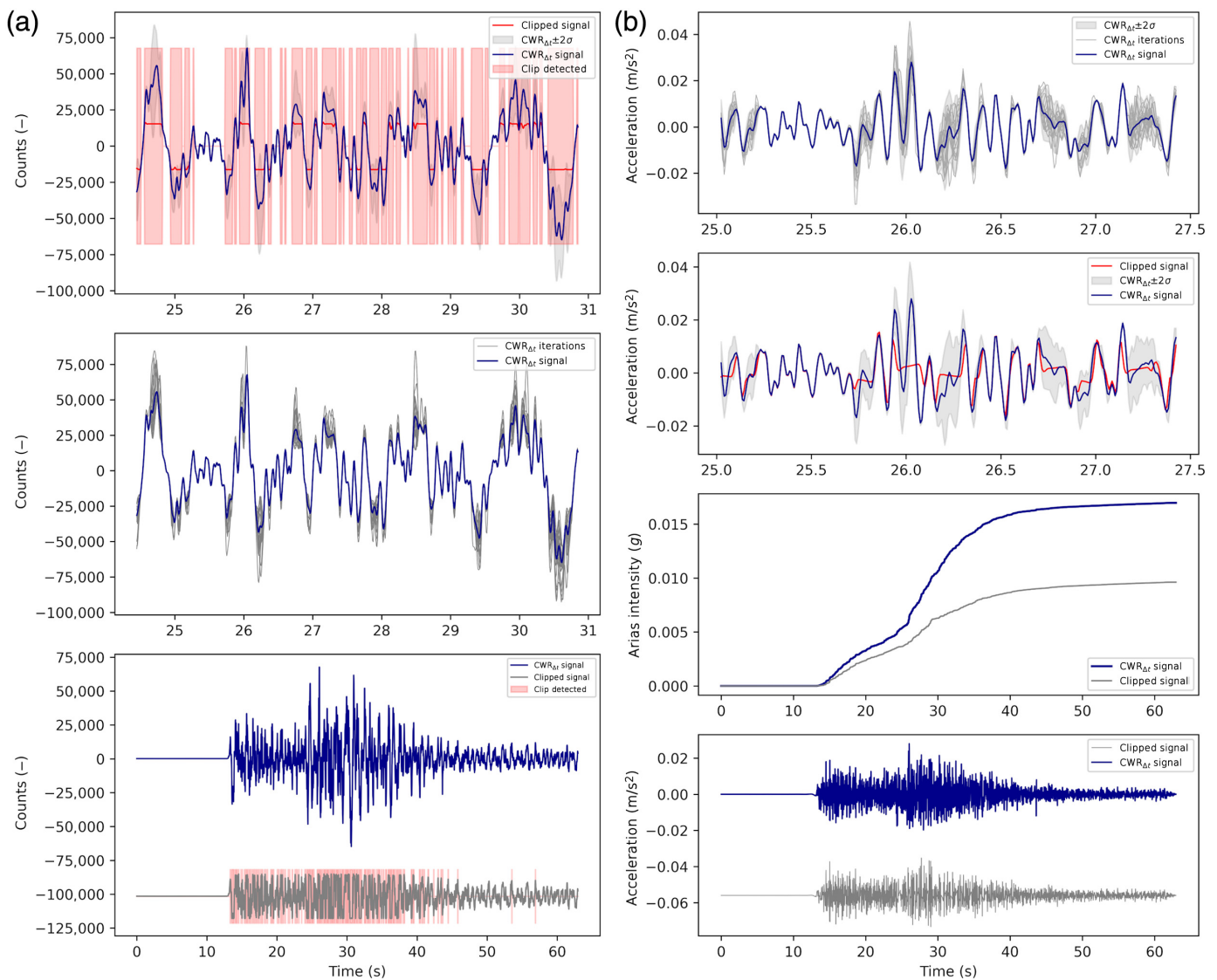


Figure 10. Record BE.CLA (single Z component) at 95 km from the epicenter. (a) Native digitizer counts (proportional to velocity), as Figure 6. (b) Acceleration-based data, as Figure 7. Detected clipping is highlighted (left top and bottom panels). The color version of this figure is available only in the electronic edition.

3%–7% and 0.7%–1.7%, respectively, at the same clipping levels. If minor clipping is suspected, it would be slightly beneficial to use the original approach, although this comes at the cost of losing the nondeterministic and nonstationary features of the $\text{CWR}_{\Delta t}$ approach. Although a comprehensive comparison with other methods has not been undertaken within the scope of this study, our conclusions support those of [Zhang et al. \(2016\)](#): both the CWR method and $\text{CWR}_{\Delta t}$ outperform traditional interpolation methods, which, for earthquake ground-motion records, quickly break down as clipping length increases beyond a few samples. Furthermore, the CWR methods are less restrictive in application than “template matching” approaches, which rely on the availability of upscaled weak-motion records (e.g., aftershocks). In addition, it should be noted that ML methods can provide complementary results with comparable performance and are useful to understand the epistemic uncertainty of reconstruction.

The CWR approaches depend on the prior identification of clipped samples. In the case of “hard clipping” due, for example,

to a limited full-scale range on low-resolution digitizers, this is relatively straightforward. Two approaches were successfully implemented to detect and identify clipped samples; one based on amplitude and one on slope detection. A challenge, beyond the scope of this article, lies with the detection and identification of “soft clipping” due, for example, to predigitization hardware issues that result in a nonlinear response of the instrument to ground motion. Although our implementation here focused on “historic” records that suffered primarily from hard-clipping due to datalogger resolution limits, existing, sophisticated detection algorithms are available (e.g., [Wang and Zhang, 2020](#)) to deal with soft-clipping on modern instruments. These clipping

detection approaches should be considered a preprocessing step to reconstruction and implemented as such alongside the CWR algorithms.

The results of the CWR _{Δt} approach can be propagated into other signal domains, such as power spectral density, Fourier spectra, and pseudodamped response spectra, providing estimated spectral amplitudes and associated uncertainties. CWR _{Δt} was benchmarked against artificially clipped empirical strong-motion records from the ESM database.

The analysis, based on the velocity (equivalent seismometer) time histories determined from those data, indicated that the method provides optimal solutions for waveform segmentation that depends on clip level. Moderate to severe clipping is significantly better reconstructed by using a spectrally dependent segment duration (defined by dominant frequency or, for earthquake records, through the proxy of magnitude). Although a targeted approach may therefore be optimal, across the range of clip levels, we find that a moderate segmentation level (defined by $c_l = 10$) in the CWR _{Δt} method leads to optimal results: at high clip levels significantly reducing bias, at a minor cost of small increases in reconstruction variability for low clip levels across all scenarios. This is useful because it is otherwise difficult to estimate clip-level a priori where no reference exists.

We applied the CWR _{Δt} method to a real-world benchmark though collocated high-/low-gain seismometers in the LDG seismic network. These data were in the form of clipped records of the M_L 5.8 Roermond earthquake along with unclipped (low-gain) reference time histories that we can compare reconstructions against. The reconstruction of the records, severely clipped at levels corresponding to 28% and 18% of true PGV, was successful, producing restored records consistent with the reference (unclipped) low-gain waveform. In addition to the recovery of peak motions (PGV and PGA), other characteristics of the waveforms were also shown to be successfully restored (such as cumulative Arias intensity, pseudospectral accelerations, and waveform coherence). On inspection, the third record (site FLN), clipped at an extreme level of 7% of true PGV, was deemed unrecoverable, leading us to suggest that the method is unsuitable for records with such extreme levels of clipping (i.e., at less than 10% of true PGV). The method nevertheless showed broad applicability to a range of earthquake scenarios (magnitudes 4.5–6.5 and distances below 150 km) in synthetic benchmark tests. We see no reason that the method would not be suitable generally to any earthquake scenario but encourage users to undertake synthetic-clipping tests on appropriate waveforms (which may be scenario- or location specific) for a given application. Such sensitivity tests are also essential to understand the most appropriate selection of setup parameters, such as segment overlap. Further work to quantitatively compare the CWR _{Δt} method with other techniques is ongoing and will be used to develop recommendations for clipped waveform reconstruction in subsequent work.

We finally applied the method to records in the epicentral region of the Roermond earthquake. We found that in addition to clipping, there was evidence of polarity reversal during clipping, which we were able to correct for prior to reconstruction. The time histories derived from these reconstructed records, both in terms of velocity and acceleration, form a valuable addition to ground-motion databases and can help improve and constrain ground-motion models for the stable continental region of northwestern Europe.

Data and Resources

Data for this work are publicly available online or via direct request: the European Strong-Motion Database is managed by Istituto Nazionale di Geofisica e Vulcanologia (INGV) and currently available online at <https://esm-db.eu/> (last accessed March 2025). The Roermond earthquake data are available on request from the network operators, KNMI (NL network), ROB (BE network), and LDG (RD network). The supplemental material to this article provides a flow-chart summarizing the approach detailed herein. The source code is openly available on request from the authors and will be made publicly available via github.com/leslieDLcy/waveclipy (last accessed December 2025).

Declaration of Competing Interests

The authors acknowledge there are no conflicts of interest recorded.

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